Distributed Image Reconstruction with Safe Multi-Agent Coordination

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***Abstract*—This project presents a multi-robot coordination framework for distributed image reconstruction. The proposed system maps image pixels to spatial target coordinates and assigns them to a swarm of non-holonomic robots. Task allocation is handled via a distributed auction-based algorithm that accommodates color-specific constraints, ensuring robots only bid on spatial points matching their pre-assigned color payload. To guarantee safe navigation during the reconstruction phase, we model the robots using unicycle dynamics and synthesize collision-free trajectories using a Control Barrier Function (CBF) and Control Lyapunov Function (CLF) formulation solved via Quadratic Programming (QP). Simulation results demonstrate the efficacy of the framework in achieving collision-free convergence to complex formations.**

***Index Terms*—Multi-Robot Systems, Task Allocation, Control Barrier Functions, Unicycle Dynamics, Swarm Robotics**

I. Introduction

Coordinating a swarm of robots to form complex, recognizable patterns requires robust solutions to both high-level task assignment and low-level trajectory generation. This project addresses the challenge of utilizing a distributed multi-robot system to physicalize digital images, effectively turning a swarm of autonomous agents into a dynamic, ”living” display. By bridging the gap between digital representation and physical multi-agent coordination, we explore the limits of swarm density and decentralized decision-making.

II. Problem Approach

The problem is decomposed into three primary stages to ensure scalability and safety in high-density environments:

1) **Spatial Coordinate Extraction:** Utilizing uniform point sampling of a source image to generate target coordinates and color.
2) **Distributed Task Allocation:** A decentralized mechanism to assign physical robots to extracted coordinates, based on distance and color.
3) **Decentralized Safety-Critical Control:** To drive the robots to their designated targets, we implement a control framework based on Control Barrier Functions (CBF). This ensures collision-free navigation even as the swarm reaches high-saturation densities at the target locations.

III. Methodology

A. Image-to-Point Mapping

Let an input image be defined as a set of pixels with corresponding RGB values. We apply Farthest Point Sampling (FPS) to extract a set of M uniformly distributed target coordinates $\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_M\}$, where $\mathbf{p}_j \in \mathbb{R}^2$. To prevent spatial conflicts at the target destinations, we enforce a minimum separation distance d_{min} such that $||\mathbf{p}_a - \mathbf{p}_b|| \geq d_{min}$ for all $a \neq b$. Each point is also quantized to a binary color representation (e.g., RGB values $\in \{0, 1\}$).

B. Color Quantization

To ensure a clean reconstruction, we treat ”white” pixels as the canvas background. A pixel is ignored if:

$$R_j > \tau_{white} \wedge G_j > \tau_{white} \wedge B_j > \tau_{white} \tag{1}$$

where $\tau_{white} = 0.95$. This significantly reduces computational overhead by focusing only on the ”content” of the image.

To facilitate multi-agent coordination, the remaining color data is quantized into a binary space. Each color channel is mapped as follows:

$$C_{binary} = \begin{cases} 1 & \text{if } C_{RGB} > 0.5 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

This quantization reduces the infinite variability of the RGB spectrum into a small, discrete set of IDs. While a binary 3-bit space allows for $2^3 = 8$ combinations, the removal of the white background $(1, 1, 1)$ leaves only 7 possible color assignments for the robots, as shown in Table I.

TABLE I
Binary Color Mapping and Robot Initialization Classes

Color ID	**Red**	**Green**	**Blue**	**Resulting Color**
1	0	0	0	Black
2	0	0	1	Blue
3	0	1	0	Green
4	0	1	1	Cyan
5	1	0	0	Red
6	1	0	1	Magenta
7	1	1	0	Yellow
8 (Ignored)	*1*	*1*	*1*	*White (Background)*

This simplification is critical for the task allocation phase; instead of robots attempting to match exact RGB values, they are pre-assigned to one of these 7 color classes. The auction algorithm then only considers pairings where $Robot_{i,color} = Point_{j,color}$.

C. Robot Dynamics

We consider a swarm of N robots. Each robot $i \in \{1, \ldots, N\}$ is modeled using non-holonomic unicycle dynamics:

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{\theta}_i \end{bmatrix} = \begin{bmatrix} \cos(\theta_i) & 0 \\ \sin(\theta_i) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix} \tag{3}$$

where $\mathbf{q}_i = [x_i, y_i, \theta_i]^T$ is the state configuration (position and heading), and $\mathbf{u}_i = [v_i, \omega_i]^T$ is the control input consisting of linear and angular velocities. The control inputs are bounded such that $v_i \in [-v_{max}, v_{max}]$ and $\omega_i \in [-\omega_{max}, \omega_{max}]$.

D. Task Allocation

To pair the N robots to the M target points, we implement a distributed auction algorithm. When color constraints are active, the auction enforces that a robot i with color ID c_i can only bid on a target point j if the point's color ID c_j matches ($c_i = c_j$).

During the bidding phase, each available robot evaluates the cost to all unassigned, color-matching targets. The cost function is strictly distance-based:

$$C_{i,j} = \alpha ||\mathbf{p}_j - \mathbf{x}_i||_2 \tag{4}$$

where $\mathbf{p}_j = [x_j, y_j]^T$ is the 2D position of the image point, $\mathbf{x}_i = [x_i, y_i]^T$ is the 2D position of the robot, and α is a scaling factor. Robots submit bids for the target that minimizes this cost. In the event of conflicting bids for the same target, the assignment is awarded to the robot with the minimum global cost, and the losing robots re-enter the bidding pool until all targets (or robots) are exhausted.

E. Control and Safety Framework

The control architecture is designed as a decentralized safety-critical system. Each agent i solves a local optimization problem to satisfy convergence to a target $\mathbf{p}_{g,i}$ while maintaining safety constraints with respect to its neighbors $\mathcal{N}_i$.

1) Control Lyapunov Function (CLF): To drive the robot to its assigned target $\mathbf{p}_{g,i} = [x_g, y_g]^T$, we define a Control Lyapunov Function (CLF) based on the Euclidean distance error $e_{pos,i} = \|\mathbf{p}_{g,i} - \mathbf{x}_i\|_2$. The desired linear velocity v_{des} is proportional to the distance, while the desired angular velocity ω_{des} minimizes the heading error α_i:

$$v_{des} = k_v \cdot e_{pos,i} \tag{5}$$

$$\omega_{des} = k_\omega \cdot \text{wrapToPi}(\text{atan2}(y_g - y_i, x_g - x_i) - \theta_i) \tag{6}$$

In implementation, a "Repulsion" term v_{repel} is added to v_{des} if a neighbor is within a critical radius to preemptively steer away before the barrier constraints become active.

2) Higher-Order Control Barrier Function (HOCBF): We define safety as a minimum separation distance D_{safe} between any two robots i and j. The safe set $\mathcal{C}$ is defined by the super-level set of the Control Barrier Function $h_{ij}(\mathbf{x})$:

$$h_{ij}(\mathbf{x}) = \|\mathbf{x}_i - \mathbf{x}_j\|^2 - D_{safe}^2 \geq 0 \tag{7}$$

To ensure the forward invariance of $\mathcal{C}$, we enforce the condition $\dot{h}_{ij} + \gamma(h_{ij}) \geq 0$. Taking the time derivative of h_{ij}:

$$\dot{h}_{ij} = \frac{\partial h_{ij}}{\partial \mathbf{x}_i} \dot{\mathbf{x}}_i + \frac{\partial h_{ij}}{\partial \mathbf{x}_j} \dot{\mathbf{x}}_j \tag{8}$$

$$\dot{h}_{ij} = 2(\mathbf{x}_i - \mathbf{x}_j)^T (\mathbf{v}_i - \mathbf{v}_j) \tag{9}$$

Substituting the unicycle dynamics $\mathbf{v}_i = [\cos\theta_i, \sin\theta_i]^T v_i$, the inequality becomes:

$$2(\mathbf{x}_i - \mathbf{x}_j)^T \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix} v_i - 2(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{v}_j + \gamma h_{ij} \geq 0 \tag{10}$$

We define the "drift" term $d_{ij} = 2(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{v}_j$ which represents the influence of the neighbor's current velocity on the barrier. The constraint is then reformulated as a linear inequality $A_{ij}\mathbf{u}_i \leq b_{ij}$:

$$\underbrace{-2(\mathbf{x}_i - \mathbf{x}_j)^T \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}}_{A_{ij}} v_i \leq \underbrace{\gamma_{local} h_{ij} - d_{ij}}_{b_{ij}} \tag{11}$$

F. Adaptive Safety Tuning

To prevent conservativeness during the final reconstruction phase, we implement an adaptive γ_{local}. As the robot approaches its goal (where $e_{pos,i} \to 0$), the safety margin is relaxed to allow for dense packing:

$$\gamma_{local} = \gamma \left(1 + \eta \left(1 - \min(1, \frac{e_{pos,i}}{e_{init,i}}) \right) \right) \tag{12}$$

where η is a scaling factor and $e_{init,i}$ is the initial distance from the robot's starting circle to the image target.

G. Quadratic Program (QP) Formulation

The safe control input $\mathbf{u}_i^*$ is found by solving the following QP at each time step t:

$$\min_{\mathbf{u}_i, \delta} \quad \frac{1}{2} \|\mathbf{u}_i - \mathbf{u}_{des}\|_W^2 + p\delta^2 \tag{13}$$

$$\text{s.t.} \quad A_{ij} v_i \leq b_{ij} + \delta, \quad \forall j \in \mathcal{N}_i \tag{14}$$

$$|v_i| \leq v_{max}, \quad |\omega_i| \leq \omega_{max} \tag{15}$$

where W is a diagonal weighting matrix emphasizing orientation correction, and δ is a slack variable that prevents the solver from failing in infeasible (high-density) configurations.

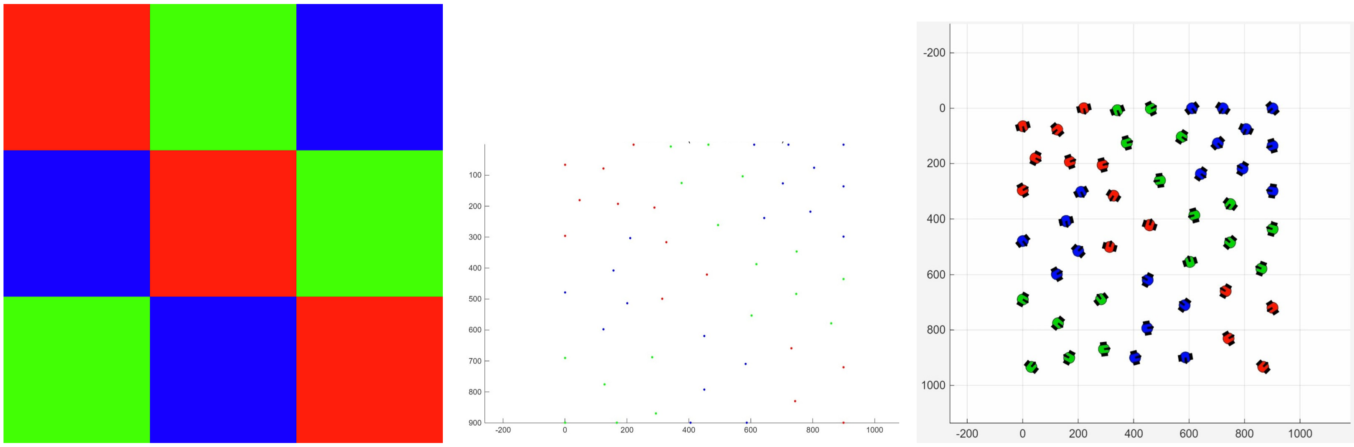

Fig. 1. The distributed image reconstruction pipeline: (Left) The original input image. (Middle) The extracted target coordinates after Farthest Point Sampling and background filtering. (Right) The final converged state of the multi-robot swarm.

H. Deadlock Mitigation: Sequential Activation and Yielding

In many-to-one or dense image patterns, robots can become trapped in local minima. Two heuristics are implemented to resolve this:

1) **Sequential Activation**: Robots are activated in "waves" based on their distance from the image center. This allows the core of the image to be populated first, reducing the probability of outer agents blocking inner agents.
2) **Directional Yielding**: If a robot i has reached its destination ($finished_i = true$) but detects a neighbor j with a significant closing rate ($\dot{h}_{ij} < 0$), it enters a "yielding" state. It temporarily abandons its goal and moves in a direction orthogonal to the neighbor's path for T_{yield} steps, clearing the collision course before returning to its target.

IV. Simulation Results

Simulations were conducted in MATLAB. The framework was tested using $N = 50$ robots tasked with reconstructing an input image.

A. Setup and Parameters

The robots were initialized in a uniform circular formation surrounding the target image coordinates. The parameters utilized were $v_{max} = 5.0$ m/s, $\omega_{max} = 6.0$ rad/s, and a simulation timestep of $dt = 0.1$s.

B. Results

The framework was evaluated by processing a source image and reconstructing it using the simulated unicycle robots. Figure 1 illustrates the three primary stages of the pipeline. First, the original digital image (left) is processed through the Farthest Point Sampling and color quantization algorithms to extract a discrete set of spatial target coordinates (middle). Finally, the distributed auction successfully allocates these targets, and the CBF-CLF control framework drives the robots from their initial circular formation to their assigned coordinates, achieving the final physical reconstruction without any inter-agent collisions (right).

C. Performance

The implementation of color-based task filtering significantly optimized the distributed auction, bounding the allocation complexity to strictly under $\mathcal{O}(N^2)$ iterations by functionally partitioning the bidding pool. During trajectory execution, the HOCBF-CLF framework robustly maintained the minimum separation distance D_{safe} across the swarm. The integration of the directional yielding heuristic and sequential activation proved critical in resolving localized minima, allowing transiting agents to safely pass through the established inner core of the formation.

However, the framework exhibited distinct operational trade-offs. While the sequential activation paradigm was necessary to prevent catastrophic gridlock at the origin, it substantially increased the time required to complete the image reconstruction. Furthermore, minor oscillatory behaviors were observed in the unicycle trajectories within densely packed regions due competing barrier constraints and the aggressive relaxation of the slack variable δ when multiple neighbors converge in a tight radius. Consequently, a direct correlation was observed between spatial density and system stability: while sparser target distributions mitigated collision risks and simplified trajectory generation, they severely degraded the visual aesthetics of the final reconstruction, particularly for images with more details. Moreover, while the color binarization process was essential for simplifying the problem and reducing the computational search space for the auction algorithm, it introduced significant discretization errors. By mapping the continuous RGB spectrum into a 3-bit space, high-frequency color variations and gradients were lost. For instance, transitional hues such as orange were systematically rounded to the nearest binary primary (red), resulting in a visible loss of chromatic detail in the final reconstruction. This highlights a fundamental trade-off between the complexity of the robot payload initialization and the visual accuracy of the generated swarm formation.

V. Conclusion and Future Work

This project successfully demonstrated a comprehensive framework for coordinating a swarm of non-holonomic robots to reconstruct digital images in a physical 2D space. By coupling a color-constrained distributed auction algorithm with a localized HOCBF-CLF control architecture, the system achieved robust task allocation and collision-free navigation into complex, multi-agent formations.

Future work will focus on addressing the identified scalability and performance trade-offs. Potential extensions include replacing the static sequential activation heuristic with a dynamic, density-aware release policy to optimize global reconstruction time without sacrificing safety. Furthermore, smoothing the oscillatory control inputs via predictive barrier formulations, exploring fully decentralized communication topologies for the bidding phase, and adapting the kinematic framework for 3D aerial vehicle (UAV) swarm coordination present highly promising avenues for continued research.